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Als or Humans in Luxury Shopping?

A Design Study of VR Shopping Assistants

Background

VR offers unprecedented opportunities for luxury retail [1].

However

The design of **shopping assistants** for replicating the consultation and expertise of human sales associates still remains underexplored.

Existing Research



Compares individual assistant types but lacks systematic evaluation across **multiple modalities** within the **same luxury shopping context** [2].

Research Question



How do different assistant types influence consumer experience in VR luxury shopping?



What design factors affect user acceptance of **webcam-based asymmetric PC-VR interaction**?



How do **human-operated** versus AI assistants differ in trust and satisfaction?



Methodology

FOUR ASSISTANT TYPES



3D Virtual Assistant

- A rigged, photorealistic character.
- Human-operated via VR controllers.**
- Capable of full mobility, gesturing, and physical product manipulation.



Webcam Assistant

- Real-time 2D video feed of a human operator displayed on a floating canvas.**
- Authentic facial expressions, but restricted to a screen.



AI Assistant

- GPT-4 driven 3D avatar.**
- Speech-to-text / text-to-speech with lip-sync.**
- Responses displayed via text overlay.



Voice Assistant

- Audio-only spatial voice chat.**
- No visual representation;
- Relies entirely on auditory cues.

TASK FLOW



STUDY DESIGN



N=20 (10M/10F, aged 24-29)
Within-subjects

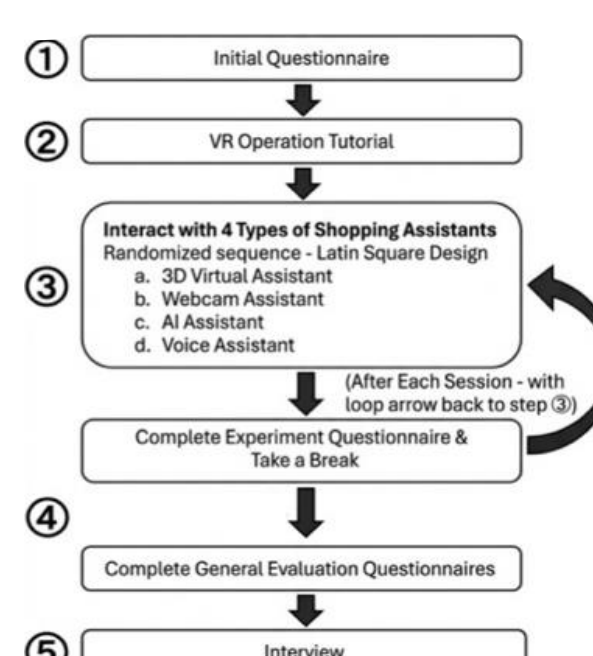


VR luxury store
(10 mins each condition)

Measures:

- NASA-TLX [3].
- 20-dimension questionnaire.
- Preference rankings.
- Semi-structured interviews.

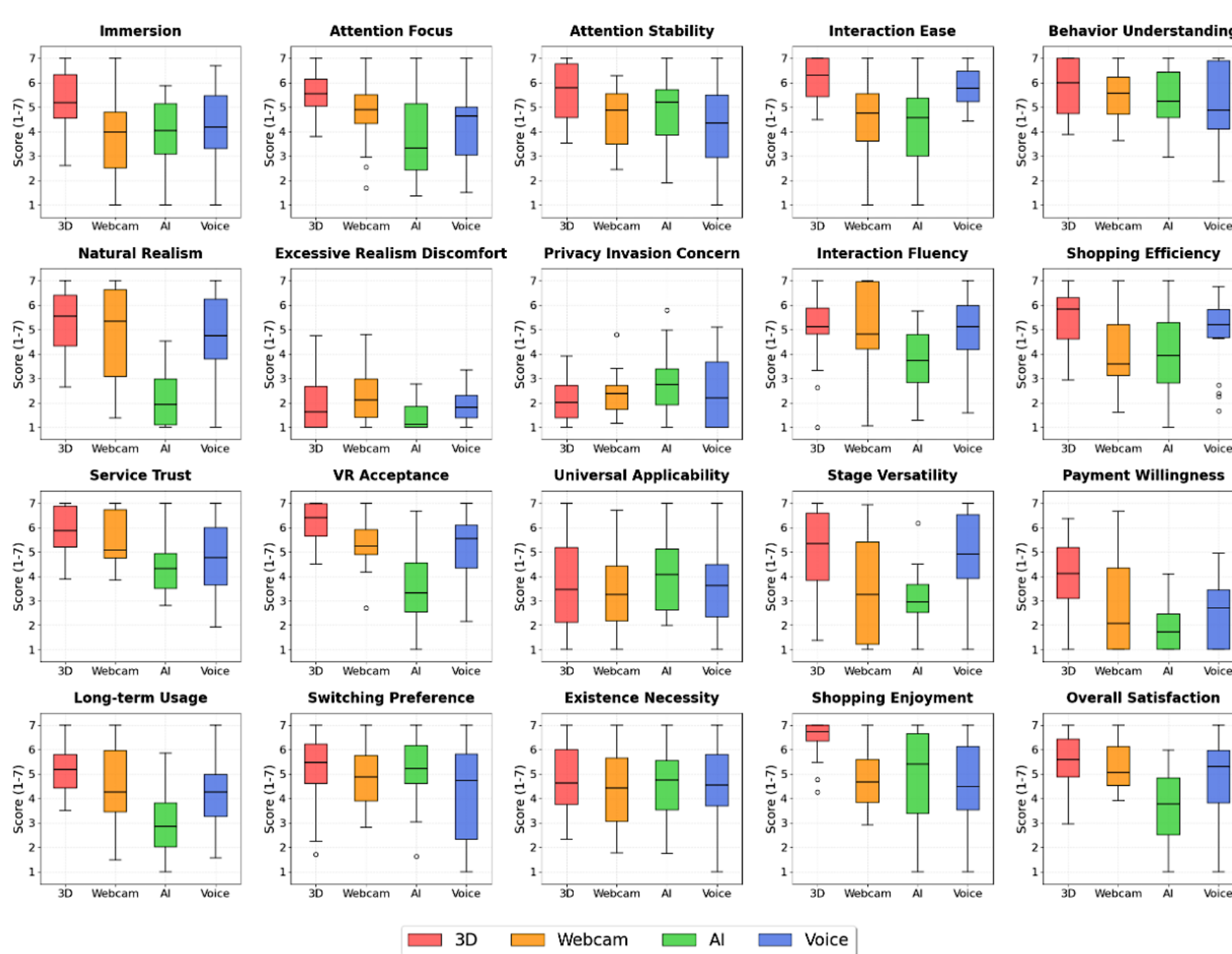
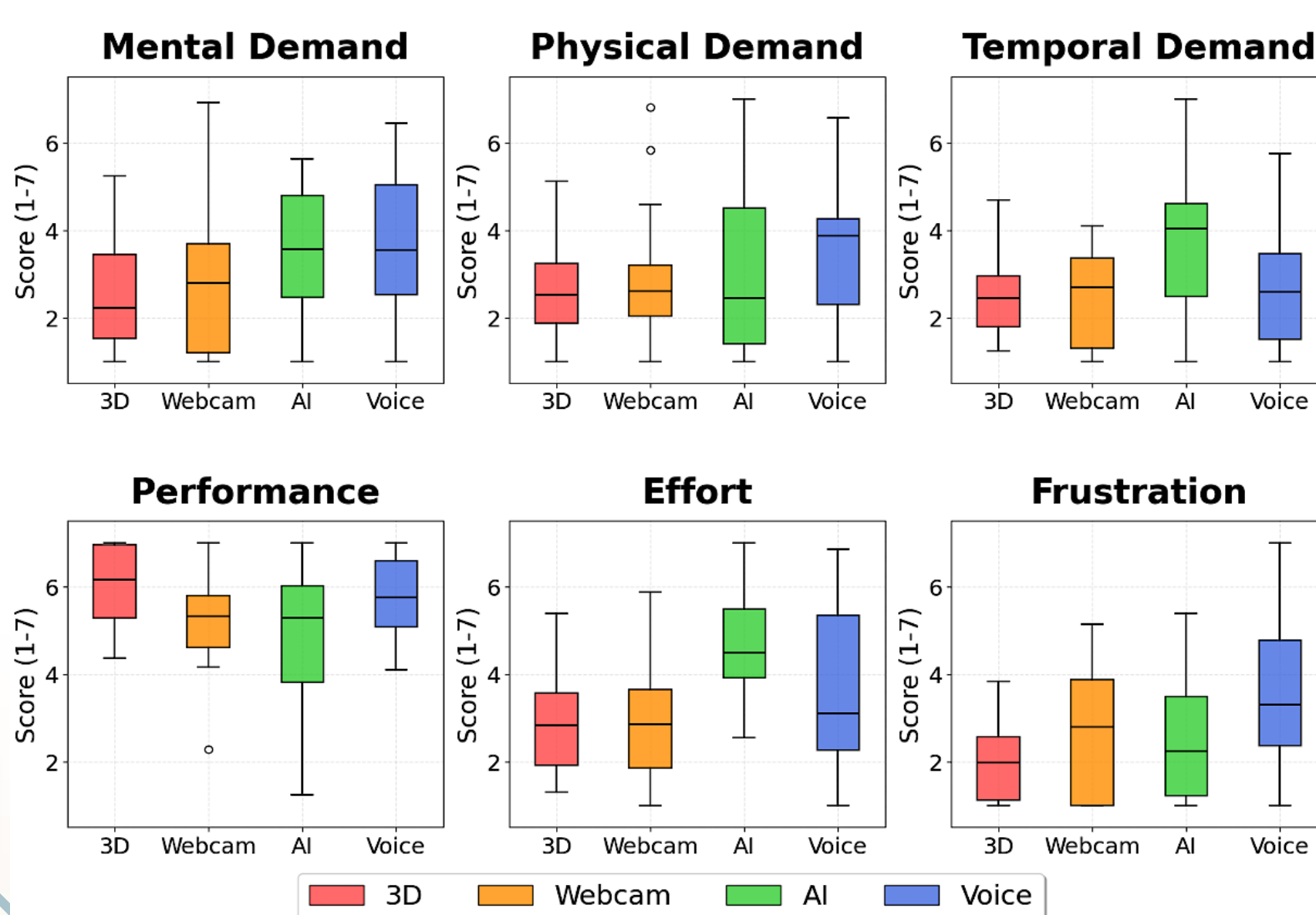
EXPERIMENTAL PROCEDURE



Results

NASA-TLX

Significant differences in Performance ($F(3,57)=2.80, p < .05$) and Frustration ($F(3,57)=2.92, p < .05$), with 3D avatars ($M=6.00$) outperforming AI ($M=4.75, p < .05$) while showing lower frustration ($M=1.90$ vs. $3.35, p < .05$)

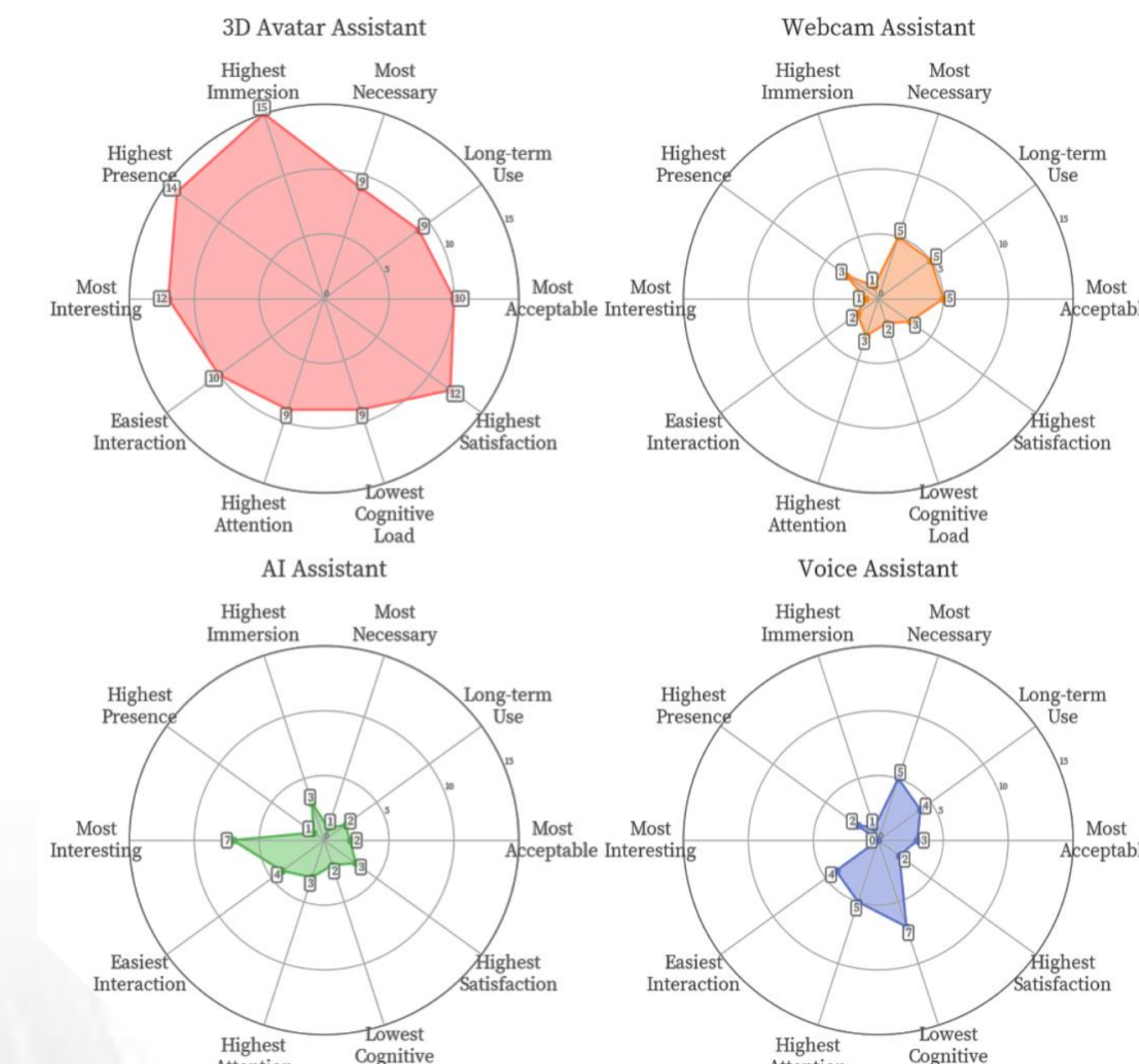


20 Evaluation Dimensions

3D avatars significantly surpassed AI in 11 metrics ($p < .05$ or $p < .01$), including Immersion, Attention Focus, Natural Realism, Service Trust, and Overall Satisfaction.

Preference Rankings

3D avatars leading with 55.5% average preference across 8/10 dimensions. Webcam achieved 50% immersion preference versus 75% for 3D avatars



Thematic Analysis

- Immersion-Realism Trade-off:** 14/20 participants experienced "dimensional misalignment" with webcam.
- Social Pressure:** 16/20 became more polite with webcam versus casual with AI assistant.
- Context Dependency:** 17/20 preferred human assistants for luxury goods.
- Trust Mechanisms:** 15/20 trusted humans for emotional support.
- Technical Limitations:** AI response delays (8/20) and webcam positioning inflexibility (12/20)

Discussion & Conclusion

Key Finding 1 Optimal Balance

Our findings demonstrate that 3D human-controlled avatars provide an optimal balance of immersion, trust, and satisfaction in VR luxury shopping environments

Key Finding 2 Immersion-Realism Trade-off

Our findings demonstrate that 3D human-controlled avatars provide an optimal balance of immersion, trust, and satisfaction in VR luxury shopping environments

Key Finding 3 Human vs AI

Human assistants (3D, webcam, voice) significantly outperformed AI in overall satisfaction ($M=5.42$ vs. 4.05), validating the critical importance of human agency for experiential products where emotional connection is paramount

Our findings demonstrate:

- 3D human-controlled avatars provide an optimal balance of immersion, trust, and satisfaction.
- The webcam condition revealed a fundamental design tension.
- Results demonstrate 3D human-controlled avatars as the optimal solution, while revealing fundamental trade-offs between realism and immersion.

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